6069 | Kerala Polytechnic Study Hub

REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Fabric Manufacture III Practical

A full-screen, syllabus-style digital handbook for Course Code 6069. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6069

COURSE TITLE
Fabric Manufacture III Practical

DEPARTMENT
Textile Technology

TYPE
Theory / Practical

SEMESTER
6

CATEGORY
Practical

USE
Study + notes PDF

FORMAT
Big handbook

6069

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

Course objectives and syllabus map

Objectives

- ✓ Operate and analyse advanced fabric manufacturing processes
- ✓ Prepare loom settings, design plan and production records
- ✓ Identify fabric defects and corrective actions

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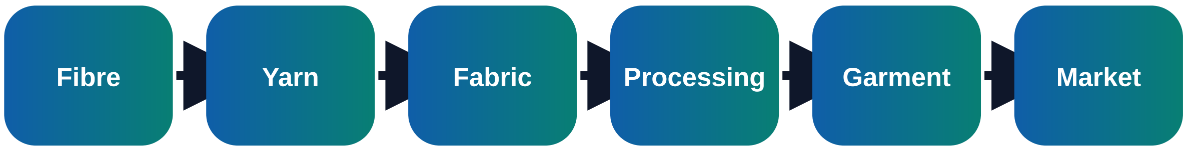
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Advanced loom preparation	Yarn selection, warp and weft preparation for required fabric; Warping, sizing idea, drawing-in, denting and gaiting	Short notes, calculations, diagram, checklist, viva answers
M2	Dobby, jacquard and design conversion	Plain, twill, satin and fancy weave planning; Dobby design, peg plan and lift plan	Short notes, calculations, diagram, checklist, viva answers
M3	Production and quality control	Pick density, ends per inch, picks per inch and cover factor; Loom efficiency, stoppages, warp break and weft break	Short notes, calculations, diagram, checklist, viva answers
M4	Practical record and cost awareness	Production calculation, fabric weight and wastage; Machine setting sheet, doffing record and shift report	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Advanced loom preparation

Core explanation

- ✓ Yarn selection, warp and weft preparation for required fabric
- ✓ Warping, sizing idea, drawing-in, denting and gaiting
- ✓ Loom timing, shedding, picking, beating-up and take-up
- ✓ Safety precautions near moving loom parts

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Dobby, jacquard and design conversion

Core explanation

- ✓ Plain, twill, satin and fancy weave planning
- ✓ Dobby design, peg plan and lift plan
- ✓ Jacquard card/pattern idea and motif placement
- ✓ Fabric design to loom setting conversion

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

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- Missing units in calculations.
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- Not connecting the topic to textile production or customer requirement.

Module 3

Production and quality control

Core explanation

- ✓ Pick density, ends per inch, picks per inch and cover factor
- ✓ Loom efficiency, stoppages, warp break and weft break
- ✓ Defects: broken end, double pick, reed mark, slub, temple mark and starting mark
- ✓ Inspection, defect marking and corrective action

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Practical record and cost awareness

Core explanation

- ✓ Production calculation, fabric weight and wastage
- ✓ Machine setting sheet, doffing record and shift report
- ✓ Fabric sample mounting and design documentation
- ✓ Power, labour and machine-hour cost idea

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Loom Efficiency % = Actual Production / Calculated Production x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

EPI = Ends per Inch

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

PPI = Picks per Inch

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Fabric Weight depends on count, EPI, PPI, crimp and area

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Prepare loom setting sheet for a selected weave
- ✓ Identify defects in fabric sample
- ✓ Calculate loom efficiency
- ✓ Prepare fabric sample record with weave diagram

Important keywords

Warp **Weft** **Dobby** **Jacquard** **EPI** **PPI** **Loom efficiency** **Fabric defect**

Short questions

Q1.1	Explain: Yarn selection, warp and weft preparation for required fabric	Answer with definition, process, application and one example.
Q1.2	Explain: Warping, sizing idea, drawing-in, denting and gaiting	Answer with definition, process, application and one example.
Q1.3	Explain: Loom timing, shedding, picking, beating-up and take-up	Answer with definition, process, application and one example.
Q2.1	Explain: Plain, twill, satin and fancy weave planning	Answer with definition, process, application and one example.
Q2.2	Explain: Dobby design, peg plan and lift plan	Answer with definition, process, application and one example.
Q2.3	Explain: Jacquard card/pattern idea and motif placement	Answer with definition, process, application and one example.
Q3.1	Explain: Pick density, ends per inch, picks per inch and cover factor	Answer with definition, process, application and one example.
Q3.2	Explain: Loom efficiency, stoppages, warp break and weft break	Answer with definition, process, application and one example.

Q3.3	Explain: Defects: broken end, double pick, reed mark, slub, temple mark and starting mark	Answer with definition, process, application and one example.
Q4.1	Explain: Production calculation, fabric weight and wastage	Answer with definition, process, application and one example.
Q4.2	Explain: Machine setting sheet, doffing record and shift report	Answer with definition, process, application and one example.
Q4.3	Explain: Fabric sample mounting and design documentation	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to Warp. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Fabric Manufacture III Practical.
2. List four important keywords: Warp, Weft, Dobby, Jacquard.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Advanced loom preparation.
7. Compare two important concepts from Dobby, jacquard and design conversion.
8. Explain the practical importance of Production and quality control in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Fabric Manufacture III Practical in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Fabric Manufacture III Practical.
2. Use keywords: Warp, Weft, Dobby, Jacquard, EPI, PPI, Loom efficiency, Fabric defect.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Advanced loom preparation

- Yarn selection, warp and weft preparation for required fabric
- Warping, sizing idea, drawing-in, denting and gaiting
- Loom timing, shedding, picking, beating-up and take-up
- Safety precautions near moving loom parts

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Dobby, jacquard and design conversion

- Plain, twill, satin and fancy weave planning
- Dobby design, peg plan and lift plan
- Jacquard card/pattern idea and motif placement
- Fabric design to loom setting conversion

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Production and quality control

- Pick density, ends per inch, picks per inch and cover factor
- Loom efficiency, stoppages, warp break and weft break
- Defects: broken end, double pick, reed mark, slub, temple mark and starting mark
- Inspection, defect marking and corrective action

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Practical record and cost awareness

- Production calculation, fabric weight and wastage
- Machine setting sheet, doffing record and shift report
- Fabric sample mounting and design documentation
- Power, labour and machine-hour cost idea

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.